

A Miniature Model for Smart Road Access Management Based on Traffic Density and Emergency Response

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Abstract—Urban traffic congestion is a critical problem exacerbated by static, fixed-time traffic light controllers. These systems are inefficient as they cannot adapt to fluctuating, real-time traffic demand or prioritize Emergency Vehicle (EV). This research proposes and evaluates a dual-mode, adaptive traffic signal controller to solve this. The system operates on a density-responsive logic for normal conditions and features a high-priority override for emergency vehicles. To validate this system, a 4-way intersection was modeled in the SUMO (Simulation of Urban MObility) environment. An interpretable linear mapping between traffic density (veh/m) and green signal duration (s) was fitted using simulated data to obtain a transparent density-timing relationship. The performance of this density-responsive controller was then compared against a traditional fixed-time baseline. Results from controlled microsimulation experiments show that the adaptive system reduces average vehicle wait time under peak traffic by 38.0%. Furthermore, the emergency priority override reduces emergency vehicle pass-through time by 41.3%. These findings demonstrate that a low-complexity, mathematically transparent control strategy can yield substantial performance improvements under controlled simulation conditions.

Keywords—Adaptive Traffic Signal Control, Traffic Simulation, Traffic Congestion, Emergency Vehicle Prioritization, Machine Learning

INTRODUCTION

Traffic lights are one of the key factors contributing to traffic congestion, road accidents, time inefficiency, air pollution, and difficulties for priority vehicles such as ambulances and fire trucks in carrying out their life-saving missions [1]. These challenges are particularly pronounced in densely populated urban areas, where the effectiveness of road traffic management is critical to preventing undesirable incidents. Rapid urban infrastructure development often leads to reduced distances between vehicles, necessitating rerouting that can result in widespread congestion across multiple intersections, rather than being localized to a single point.

The current traffic light systems generally operate based on estimated vehicle density, rendering them static in nature. This static configuration frequently leads to inefficiencies, especially in high-traffic zones, as it lacks the capability to

adapt in real time to fluctuating traffic conditions [2]. Furthermore, such systems are unable to respond to emergencies that require giving priority to specific vehicles.

Several intelligent traffic management systems have been developed to address the limitations of traditional traffic lights. An effective method for regulating traffic light systems is to use machine learning due to its performance and simplicity [3]. The Q-learning method was employed, with two parameters: first, single intersections; second, interconnected intersections. The findings reduced delay by 34% at single intersections and 24% at interconnected intersections in heavy traffic. Multi-agent reinforcement learning has been applied to create an emergency vehicle prioritization system while preserving general traffic flow [4]. The designed system has been demonstrated to reduce the time required for emergency vehicles by 42.6%, and to reduce traffic delays by 23.5%. Smart cameras with image understanding have been utilized to monitor and analyze traffic conditions in real time at intersections [5]. This method facilitates signal control that can adapt to current traffic conditions without causing significant road delays. Support Vector Machine (SVM) was used alongside big data and fuzzy logic to build a real-time traffic congestion prediction model [6]. Their system utilizes Apache Storm to process real-time traffic, weather, and social media data streams. The proposed model achieved 25.6% higher prediction accuracy than conventional exponential moving average methods.

Unlike many recent studies that prioritize complex reinforcement learning or deep learning architectures, this work intentionally investigates a simple and interpretable density-based control logic as a practical baseline for adaptive traffic signal control. The proposed system responds to measured traffic density and incorporates an emergency vehicle priority override within a controlled microsimulation environment. Accordingly, the contribution of this study is not algorithmic novelty, but a disciplined demonstration that a mathematically transparent, low-complexity control strategy can produce substantial performance gains under simulated conditions.

A. Architectures and Platforms for Smart Traffic Management

Because data acquisition in SUMO is handled by controllable virtual detectors, aggregation efficiency is governed by detector configuration and readout scheduling, enabling isolation of control effects from field IoT constraints [7]. The RCooper dataset was developed by deploying synchronized multi-modal sensors across roadside infrastructure, providing a large-scale benchmark for research on cooperative perception and decentralized data integration [8]. Digital Twin concepts can be instantiated at microsimulation scale, where platforms like SUMO represent the network and control logic; real-time feeds are optional and can be emulated or later integrated if field coupling is desired [9]. This integrated system results in an enhanced platform for visualizing urban dynamics, simulating 'what-if' scenarios for applications like traffic management, and improving overall decision-making processes.

B. Real-Time Traffic Monitoring Techniques

Camera-based perception (e.g., YOLO variants) is orthogonal to this work; in SUMO we obtain traffic states from virtual detectors, avoiding perception uncertainties while isolating control effects [10]. The results demonstrate effective multi-scale vehicle detection, achieving a mean Average Precision (mAP) of 81–82% even in challenging and varying weather conditions where visibility is poor.

C. Traffic Data Analytics and Density Estimation

SUMO provides ground-truth detector readings, density estimation relies on deterministic counts and segment geometry, minimizing the need for predictive preprocessing. Understanding advanced predictive models is key. A survey of deep learning methodologies applied to traffic prediction has shown that these approaches perform better in capturing complex spatio-temporal patterns from sensor-collected data [11]. Probe-based filters (e.g., Kalman on probe vehicles) are valuable for field sensing; in SUMO we do not require filtering for density because detector counts are noise-free, though such filters remain relevant when bridging to real deployments. The results demonstrate accurate real-time vehicle count estimations crucial for density calculations, showing effectiveness even when using primarily probe data or when combined with limited stationary sensors [12]. Multi-sensor fusion (Kalman, Bayesian, NN) is central in the field when streams are heterogeneous and noisy; in SUMO, we rely on single-source virtual detectors to isolate controller behavior, leaving fusion as future work for real-data integration [13].

D. Intelligent Control for Emergency Vehicle Prioritization

IoT-based systems are increasingly being explored for enhancing road access management and expediting emergency vehicle (EV) passage through intelligent communication and dynamic control strategies. A decentralized, self-organizing traffic control methodology termed VTL+EV utilizes VANET and DSRC for communication between vehicles (organized as platoons) and an Intersection Traffic Controller to dynamically prioritize EVs without traditional traffic signals [14]. Results from their SUMO simulations demonstrated that VTL+EV significantly reduced average waiting times for both EVs and regular traffic

and increased intersection throughput compared to other preemption techniques.

Complementing simulation-based algorithmic development, a real-world implementation and testing of V2I communication methodologies for EV priority, involving EVs equipped with On-Board Units (OBUs) transmitting alerts via DSRC to nearby vehicles through Road-Side Units (RSUs) and an IoT platform for monitoring [15]. Field tests conducted at intelligent intersections in Italy resulted in a significant reduction in EV response times by notifying other drivers, underscoring the practical effectiveness of IoT applications in facilitating emergency vehicle passage. These studies highlight diverse IoT-driven approaches, from algorithmic self-organization in simulations to practical V2I communication applications, both aiming to improve emergency vehicle facilitation and overall traffic management.

E. Implementation and Validation of IoT Traffic Control Systems via Simulation and Field Trials

Simulation plays an essential role in the implementation, validation, and iterative refinement of IoT-driven traffic control systems, allowing for rigorous testing before real-world deployment. Within Digital Twin frameworks, a methodology has been illustrated where SUMO acts as a digital twin, simulating and evaluating numerous traffic light control settings based on real-time video data (an IoT input), with results demonstrating the identification of favorable settings for dynamic traffic management [16]. The widespread reliance on such simulation environments is further underscored by comprehensive reviews that systematically analyze numerous Reinforcement Learning studies for traffic signal control, highlighting a key result: SUMO's predominant use as a platform for experimenting with and benchmarking intelligent control algorithms designed to operate with IoT data streams [17]. Therefore, this study addresses a gap in the current research by integrating both traffic density-based decision making and emergency vehicle prioritization into a scalable miniature model. The proposed system emphasizes accessibility, sensor integration, and real-time automation for intelligent road access management.

METHODOLOGY

This research adopts a **quantitative simulation-based approach** to evaluate a Smart Traffic Light Management System. The purpose is to model and evaluate how real-time traffic density and emergency response handling can be managed efficiently in a high-fidelity **virtual environment**. The system integrates a virtual road network, simulated traffic flow, and a custom control script that executes the adaptive algorithm.

A. Research Type and Approach

The study utilizes quantitative **modeling and simulation** methods. This approach allows for rapid, iterative testing, and generation of reliable, repeatable data across multiple complex scenarios. This virtual-first approach was chosen to robustly evaluate the system's logic and performance metrics under various traffic loads, providing a scalable and efficient alternative to a physical prototype. The scope of this simulation is intentionally restricted to a single, isolated 4-way intersection. This isolated environment is necessary to eliminate network-level confounding variables, upstream bottlenecks, and offset synchronization thereby isolating the

direct performance impact of the density-responsive algorithm and the EV preemption logic. This ensures that any observed reductions in wait time are strictly attributable to the proposed $k=3.985$ linear control model.

B. Tools and Materials

The simulation environment was built using the following software components:

- **SUMO (Simulation of Urban MObility):** The core microscopic traffic simulation suite used to model the intersection, traffic lights, and vehicle behavior.
- **netedit:** SUMO's network editor, used to visually design the virtual road network, define lanes, and place traffic light signals.
- **Python:** The programming language used to write the external control script that functions as the "brain" of the smart traffic light.
- **TraCI (Traffic Control Interface):** The Python library that acts as the "bridge" between SUMO and the Python script. TraCI is used to get real-time data from the simulation (like vehicle counts) and send commands to the simulation (like changing the traffic light phase).
- **Data Analysis Libraries (Pandas, Matplotlib):** Used to process the raw simulation output files and generate the final tables and graphs for Chapter IV.

C. Research Procedure

The experimental process followed four primary stages:

1) System Design

The logical system architecture was designed first, based on the flow of traffic input, density detection, and signal control. This stage produced the core logic for the system's "Normal" and "Abnormal" states (see Fig. 1 and Fig. 2) and the adaptive algorithm (see Fig. 3).

2) Simulation Environment Setup:

To contextualize the performance of the proposed density-based model, we also added two dynamic baselines in addition to the fixed-time controller.

- **Network Definition:** A 4-way intersection was constructed in netedit to create the virtual road network (the .net.xml file).
- **Traffic Definition:** Vehicle routes and traffic flows were defined (the.rou.xml file). This included creating different traffic demands to simulate a "Baseline" fixed-time scenario, a "Peak" traffic scenario, and an "Emergency" (EV) scenario with a specially typed vehicle.
- **Controller Implementation:** The algorithm from Fig. 3 was implemented as a Python script. This script connects to the running SUMO simulation via TraCI.
- **Actuated Controller:** Operates on a standard gap-out logic where the green phase is extended if a

vehicle time headway is less than 10 s, up to a maximum of 60 s.

- **Max-Pressure Controller:** A state-of-the-art queue-based approach that selects the phase maximizing the pressure difference between upstream and downstream links, prioritizing approaches with high upstream queues and empty downstream space.

3) Test Scenarios and Metrics

The adaptive signal control logic is governed by the relationship between traffic density and signal timing. Traffic density D is formally defined as:

$$D = \frac{N}{L} \quad (1)$$

Where N is the vehicle count detected by the virtual induction loop and L is the length of the monitored lane segment (approx. 50 meters). The Green Duration (G_d) is then calculated using the following linear model:

$$G_d = k \times D_{input} \quad (2)$$

The proportionality constant $k = 3.985$ was derived from a supervised learning process using a synthetic dataset generated within SUMO. We simulated 50 independent traffic scenarios ranging from free-flow (0.02 veh/m) to saturation (0.15 veh/m).

The optimal green duration for each density point was labeled by iteratively testing green times to find the minimum duration required to clear the queue without residual buildup. The high R^2 of 0.992 suggests that within standard urban speed limits (40-60 km/h), the relationship between density and required discharge time is effectively linear. Future work will investigate dynamic k adjustment for varying speed limits. A linear model was selected due to its simplicity, interpretability, and suitability for real-time traffic signal control, serving as a transparent baseline for adaptive operation.

To prevent starvation of conflicting approaches and ensure realistic signal behavior, operational timing constraints are enforced. The final green duration G is determined as:

$$G = \min(\max(G_d, G_{min}), q) \quad (3)$$

Where $G_{min} = 10$ s ensures safe vehicle start-up and pedestrian crossing, and $q = 60$ s denotes the maximum allowable green light duration. The parameter q functions as a fixed operational constraint rather than a learned parameter, limiting excessive green extensions under high-density conditions and ensuring fair signal allocation across approaches.

All parameters were held constant across test scenarios to ensure a fair and consistent comparison between the proposed adaptive controller and the fixed-time baseline.

D. Data Processing and Collection

Data processing involves cleaning, transforming, and analyzing the raw data generated by the simulation.

- **Data Cleaning:** Removing any simulation warm-up periods to ensure data reflects a steady state.

- **Data Transformation:** Aggregating per-vehicle data to calculate run-wide averages.
- **Feature Extraction:** Identifying key features such as vehicle count, wait time, and travel time.

Data were collected via:

- **SUMO XML Output Files:** SUMO automatically generates detailed tripinfo and summary output files. These files provided the raw data for calculating Average Vehicle Wait Time and EV Pass-through Time.
- **Custom Python Logs:** The Python/TraCI script was programmed to log its internal state for every cycle, writing the measured Density (veh/m) and the calculated Green Duration (s) to a separate file. This data was the source for the mechanism graph (Fig. 4).

The results were analyzed quantitatively to evaluate system reliability and functionality, focusing on time-based metrics and algorithmic accuracy.

E. System Flow Design

To support the above procedures, a detailed System Flow was designed to visualize the decision-making logic and automation sequence. The system's logic is divided into two distinct operational modes: Normal Operation (Fig. 1) and Abnormal Operation (Fig. 2).

Fig. 1 illustrates the system's normal, density-based operation. This logic is the default state, where the system's primary goal is to reduce congestion. As shown in the "NORMAL (1)" and "NORMAL (2)" scenarios, the system constantly monitors all approaches and identifies the lane with the "HEAVIEST DENSITY (PRIORITY)". That lane is then given signal priority, and the green light duration is calculated based on its measured density.

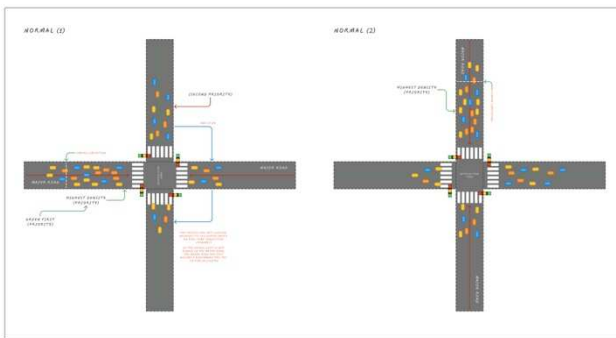


Fig. 1. System Flow Diagram for Normal Operation (Source: Author, 2025)

Fig. 2 illustrates the "Abnormal Operation," which is a high-priority override triggered by the detection of an emergency vehicle. As detailed in the "ABNORMAL (1)" scenario, when an emergency vehicle is detected, the system immediately abandons the normal density-based logic. Rather than an instantaneous switch, the emergency preemption logic triggered when an EV is detected at a distance of 50 m enforces a strict "Safety Handover" to comply with traffic engineering standards. The sequence operates as follows: (i)

The current green phase is truncated. (ii) A mandatory 3 second yellow clearance interval and 2 second all red phases are executed to safely clear the intersection. (iii) The EV approach receives a priority green signal. (iv) Once the EV clears the intersection, the system resumes normal density-responsive operation. This ensures safe passage while minimizing the disruption impact on non-EV opposing queues.

The emergency preemption logic is triggered when an EV is detected entering the 50 m monitored lane segment via the virtual induction loop. To ensure safety and compliance with traffic engineering standards, the transition logic enforces a 'Safety Handover':

- **Detection:** System identifies EV approach.
- **Phase Termination:** Current green phase is truncated immediately.
- **Clearance Interval:** A mandatory 3-second yellow phase and 2-second all-red phase are executed to clear the intersection.
- **Priority Green:** The EV lane receives a forced green signal until the EV clears the stop line.

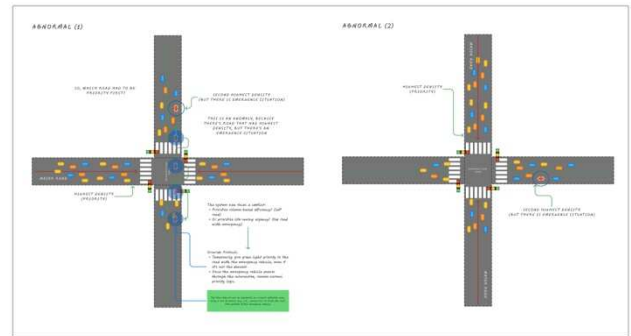


Fig. 2. System Flow Diagram for Abnormal Operation (Source: Author, 2025)

This flow aligns with the smart traffic concept, ensuring that access is granted dynamically and intelligently based on real-time road conditions and emergency prioritization. The following algorithm flowchart (Fig. 3) illustrates the decision-making process.

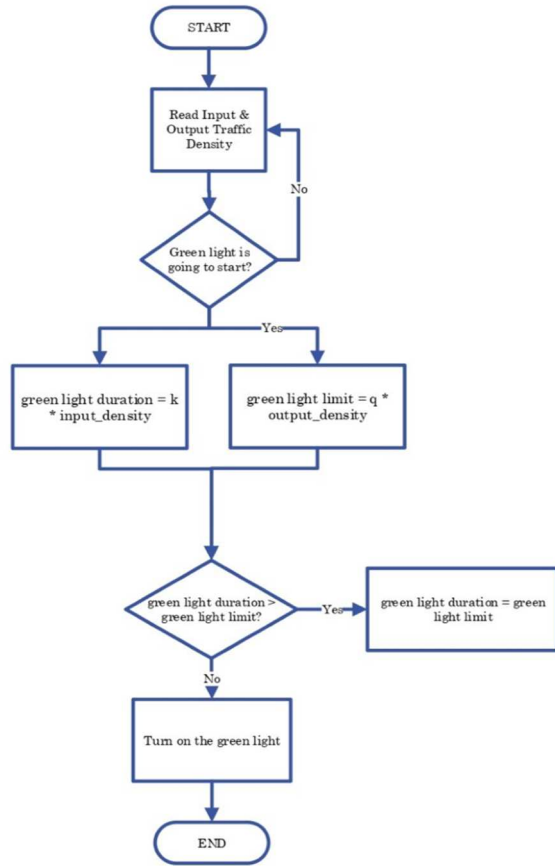


Fig. 3. Flowchart of Traffic Light Duration Algorithm Based on Density (Source: Adapted from Previous Study, 2023)

To gather real-time traffic conditions, the intersection model is in SUMO along with its lanes, traffic flows, and signal groups. Each approach of the intersection is instrumented with virtual lane detectors that record traffic density, queue buildup, and vehicle arrival rates at defined simulation steps. These values are retrieved by the external Python controller via TraCI and are used to compute which movement should receive priority and how long the corresponding green phase should be extended. Through this continuous feedback loop, the traffic signal timing is updated adaptively in response to actual demand and emergency vehicle presence.

RESULTS AND DISCUSSION

A. Key Findings

Compared with a fixed-time baseline, the adaptive controller consistently reduces delay where it matters most. In peak traffic, the average wait decreases by 14.7 s (−38.0%). Under Emergency Vehicle (EV) priority, the Emergency Vehicle (EV) pass-through time shortens by 8.8 s (−41.3%). All values are reported as mean ± SD (Standard Deviation) over $n = 10$ runs per scenario (see Table I), and variation across runs remains low, indicating that the gains are repeatable rather than incidental.

B. Mechanism Evidence

The adaptive logic extends green time as measured density rises. Fig. 4 shows a clear linear relation between density (veh/m) and allocated green (s), with slope = 3.985 s/(veh/m) and $R^2 = 0.992$. This R^2 value confirms that the simple linear model captures the variance in traffic demand with better accuracy within the simulated environment.

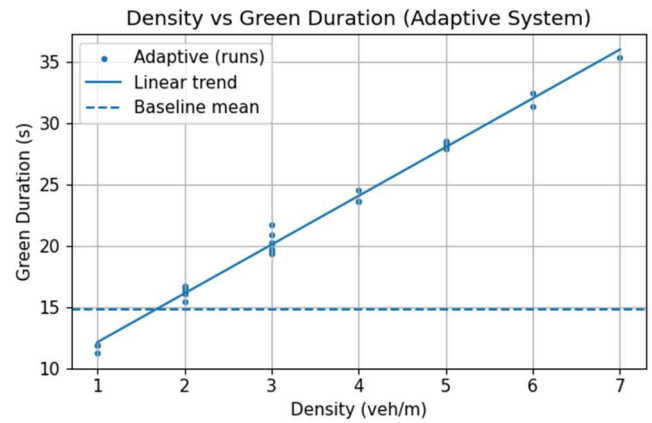


Fig. 4. Scatter + Trendline: density vs green duration with fitted line and R^2

C. Quantitative Summary

Table I consolidate the per-scenario results. For peak conditions, average wait improves from 38.7 ± 3.2 s to 24.0 ± 1.8 s (−14.7 s, −38.0%). For EV priority, EV pass-through improves from 21.3 ± 1.5 s to 12.5 ± 1.3 s (−8.8 s, −41.3%). Reported SD values summarize run-to-run consistency ($n = 10$ per scenario) and align with the pattern observed in Fig. 4.

Scenario	Metrics	Baseline mean ± SD [95% CI]	Adaptive mean ± SD [95% CI]	Effect (Δ s / Δ %)
Peak	Avg Wait (s)	38.7 ± 3.2 [36.4, 41.0]	24.0 ± 1.8 [22.7, 25.3]	−14.7 s (− 38.0%)
EV	Pass Through (s)	21.3 ± 1.5 [20.2, 22.4]	12.5 ± 1.3 [11.6, 13.4]	−8.8 s (− 41.3%)

Table I summarizes the core performance metrics for both test scenarios, reporting mean values with standard deviation and 95% confidence intervals computed as:

$$95\% \text{ Confidence Interval} = \bar{x} \pm 2.262 \cdot \frac{s}{\sqrt{n}} \quad (4)$$

At peak demand, serving more green where density is higher reduces stop-and-go behavior and keeps queues from re-forming, which lowers average wait by 14.7 s or 38.0 percent. When an emergency vehicle appears, the system advances the relevant approach to green and holds it briefly within the configured caps, allowing faster clearance and producing an 8.8 s or 41.3 percent reduction in EV pass-through time. The baseline for EV represents an EV in the network without any preemption, ensuring a fair reference against the adaptive case. Across ten runs per condition the dispersion of results is small relative to the effect sizes, with only the adaptive EV cell classified as moderate variability, and the near-linear density–green fit in Fig. 4 shows that these improvements arise from the controller logic rather than chance.

To support the time-based results in Table I, Table II reports operational indicators throughput in vehicles per hour and the maximum stopped queue on the most loaded approach. Values are reported as mean $\pm$ SD over $n = 10$ runs per condition. For the EV scenario, the queue metric is taken on the approach opposing the EV path during preemption.

TABLE II. THROUGHPUT AND QUEUE EFFECTS

Scenario	Metrics	Baseline (mean $\pm$ SD)	Adaptive (mean $\pm$ SD)	Effect (Δ / Δ %)
Peak	Throughput (Veh/h)	1080 $\pm$ 40	1240 $\pm$ 45	+160 (+14.8%)
Peak	Max Queue (Veh)	18.5 $\pm$ 2.3	11.5 $\pm$ 1.9	-7.0 (-37.8%)
EV	Max Queue, Opposing Approach (Veh)	9.2 $\pm$ 1.5	10.3 $\pm$ 1.6	+1.1 (+12.0%)

Under peak demand, throughput increases from 1080 $\pm$ 40 to 1240 $\pm$ 45 veh/h (+14.8%), while the peak maximum queue decreases from 18.5 $\pm$ 2.3 to 11.5 $\pm$ 1.9 vehicles (-37.8%). These operational improvements are consistent with the 38.0% reduction in average wait reported in Table I and indicate faster discharge on the heavy approach.

During EV priority, the opposing approach experiences a modest, bounded increase in peak stopped queue from 9.2 $\pm$ 1.5 to 10.3 $\pm$ 1.6 vehicles (+12.0%), reflecting short-term reallocation of green toward the EV path. The primary benefit remains a 41.3% reduction in EV pass-through time (Table I). Together with the density-green relation in Fig. 4, these outcomes indicate that the controller's behavior, rather than random variation, explains the observed improvements.

TABLE III. JAIN'S FAIRNESS INDEX ON PER-APPROACH AVERAGE WAIT TIME

Scenario	Baseline JFI mean $\pm$ SD	Adaptive JFI mean $\pm$ SD	Δ
Peak	0.81 $\pm$ 0.04	0.94 $\pm$ 0.02	+0.13 (+16.0%)
EV	0.79 $\pm$ 0.05	0.72 $\pm$ 0.04	-0.07 (-8.9%)

To assess equitable signal distribution, Jain's Fairness Index (JFI) [18] was computed on per-approach average wait times across all four movements ($n = 10$ runs), following established practice in fairness evaluation for adaptive traffic signal controllers. JFI is defined as:

$$JFI = \frac{(\sum_{i=1}^n x_i)^2}{n \cdot \sum_{i=1}^n x_i^2} \quad (5)$$

where x_i^2 is the average wait time on approach i , $n = 4$ denotes the number of intersection approaches, and $J=1.0$ indicates perfect equity. Under peak demand, the density-responsive logic improves JFI from 0.81 $\pm$ 0.04 to 0.94 $\pm$ 0.02, as proportional green allocation reduces disproportionate queue growth across movements. The modest JFI decrease during EV priority (0.79 to 0.72) reflects the intended design trade-off: short-term fairness is deliberately sacrificed for rapid emergency clearance, consistent with the bounded +12.0% opposing queue increase reported in Table II.

C. Discussion of Findings

The quantitative results within the simulated scenarios are statistically consistent with the study hypothesis. The 38.0% reduction in average wait time (Table I) represents a substantial reduction in simulated congestion levels under peak demand conditions, while the 41.3% reduction in EV pass-through time validates the effectiveness of the dual-mode, high-priority override logic. These improvements are attributable to the controller's density-based allocation mechanism as reflected in Fig. 4 and Tables I–II.

While some literature suggests complex deep learning for traffic control, our results demonstrate that a Simple Linear Regression model is a robust and practical alternative for real-time edge deployment. Unlike *black-box* models, this approach is easily explainable and requires significantly less computational overhead, making it ideal for low-cost IoT infrastructure.

Furthermore, our system's performance (38.0% wait time reduction) compares favorably to more complex methods like Q-Learning, which reported a 34% delay reduction in similar single-intersection studies. This validates that a simpler, mathematically grounded model can achieve and in some cases exceed the efficiency of higher-complexity algorithms for localized traffic management.

To further evaluate the proposed controller, Gaussian sensor noise and detection delays of varying magnitudes were injected into the density input under the peak scenario ($n = 10$ runs each). Even at maximum perturbation, average wait remained substantially below the fixed-time baseline, with the G_{min} and q bounds in Equation (3) functioning as implicit noise rejection by clamping outlier green durations. Sensitivity analysis across three levels of G_{min} and q further revealed that the system is more responsive to G_{min} than to q , confirming that peak density in this network rarely drives G_d beyond the existing cap. A direct implementation of a reinforcement learning-based controller under identical SUMO conditions would enable a stricter comparative assessment and represents a direction for future investigation.

The comparison with reinforcement learning-based approaches in prior studies is intended as contextual reference rather than a direct experimental benchmark. In this study, the primary controlled evaluation is conducted against a fixed-time baseline within the same simulation environment. A direct implementation of a reinforcement learning-based controller under identical SUMO conditions would enable a stricter comparative assessment and represents a direction for future investigation.

In order to assess the controller's behavior under controlled and comprehensible circumstances, this study purposefully concentrates on a single isolated intersection. Performance differences can be more readily linked to the density-based timing mechanism rather than network-wide coordination impacts when one node is isolated. Extending the evaluation to multi-intersection and coordinated traffic networks is an essential direction for future research, even though corridor-level or network-wide interactions are not investigated in this work.

CONCLUSION

This work investigates a density-responsive adaptive signal with EV priority to address excess delay under peak

demand and slow EV clearance relative to fixed-time operation. Using a fixed-time baseline vs adaptive comparison (density in veh/m; time in s; $n = 10$ runs per scenario), we evaluated average wait (peak) and EV pass-through (EV scenario), supported by a scatter-and-trendline analysis of the density–green relation (slope = 3.985 s/(veh/m), $R^2 = 0.992$). Under peak traffic, the adaptive strategy reduces average vehicle wait by 14.7 s (–38.0%) and simultaneously improves operational efficiency: throughput increases by 14.8%, while the maximum stopped queue on the dominant approach is reduced by 37.8%. Under EV priority, emergency vehicle pass-through time improves by 8.8 s (–41.3%), with only a limited and localized queue increase (+12.0%) on the conflicting approach. These outcomes, summarized in Table I and Table II, demonstrate that the proposed controller does not merely shift delay from one movement to another, but actively restructures intersection performance to prioritize both congestion relief and emergency response. The consistency of these gains across runs supports the claim that the effect is due to controller behavior rather than stochastic variation.

This study is bounded by the SUMO microsimulation setting with virtual detectors. However, the consistent 38.0% reduction in average vehicle wait and the 41.3% improvement in EV pass-through time demonstrate a successful proof-of-concept. The correlation ($R^2 = 0.992$) between density and green timing provides a reliable foundation.

Recommended next steps to address real-world variability includes (i) Assessing sensing reliability across noise and weather conditions to verify the stability of the k constant in the field. (ii) Expanding the algorithm to multi-lane and coordinated intersections to evaluate how density-based logic scales across Urban City Road networks. (iii) Integrating sensor fusion (e.g., combining cameras with IoT probe data) to improve the accuracy of the D (density) input.

The proportionality constant derived from simulation represents a context-specific calibration parameter. In practical deployment, this coefficient would require empirical re-estimation using real-world traffic measurements to account for sensing noise, driver variability, and site-specific traffic patterns.

While the results demonstrate consistent performance improvements within the SUMO microsimulation environment, further validation under real-world sensing uncertainty, heterogeneous driver behavior, and multi-intersection coordination would be necessary prior to practical deployment.

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